

Deep Space Autonomous Navigation

Page 1: X-ray Pulsar Navigation (XNAV)

XNAV utilizes periodic periodic millisecond X-ray pulsars as celestial galactic clocks. Spacecraft measure pulse times-of-arrival (TOA) to compute autonomous position vectors accurate to within 5 kilometers anywhere in the solar system.

Deep Space Network Antenna Systems

Page 2: 70-Meter Parabolic Reflectors

NASAs Deep Space Network operates three 70-meter dish complexes in Goldstone California, Madrid Spain, and Canberra Australia. They utilize cryogenic cooled ruby maser amplifiers providing communications across interplanetary distances.